KNN & K-Means: The Essential Visuals

20 Charts Every Data Scientist Should Know

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Methods and Algorithms — MSc Data Science

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The Machine Learning Pipeline

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Learning Objectives

After this lecture, you will be able to:

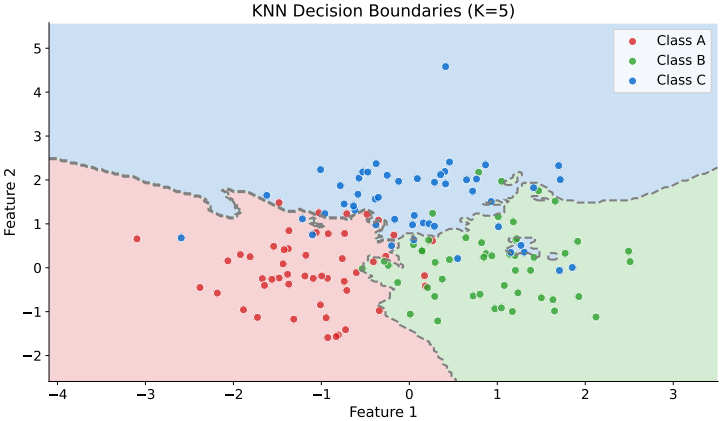
- **Interpret** KNN and K-Means visualizations to diagnose model behavior
- **Identify** when algorithms succeed and when they fail using diagnostic charts
- **Select** optimal hyperparameters (k, $n_clusters$) using visual tools

We cover 20 essential visualizations spanning KNN foundations, K-Means clustering, and advanced diagnostics

Outline

Three parts: KNN Foundations, K-Means Clustering, Advanced Diagnostics

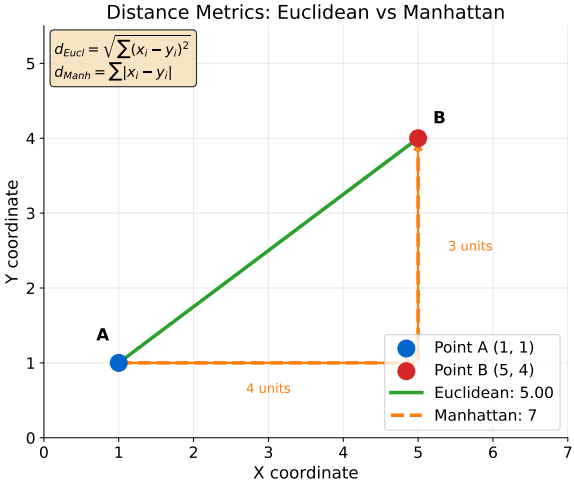
How KNN Carves Up Feature Space



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L03_KNN_KMeans/01_knn_boundaries

KNN creates flexible, non-linear boundaries that adapt to local data structure

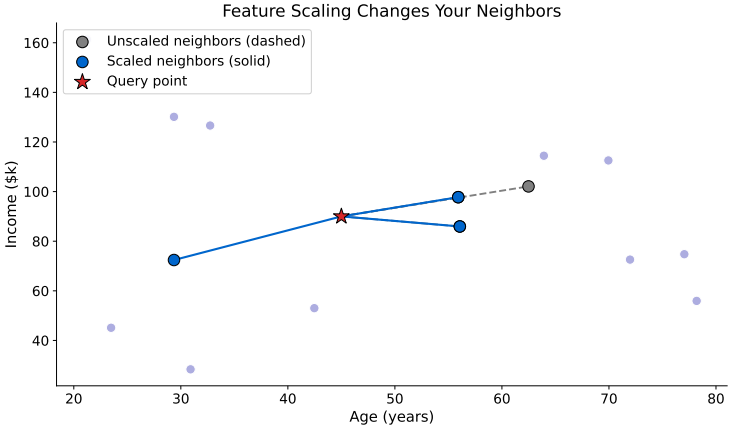
Distance Metrics: The Foundation of KNN



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L03_KNN_KMeans/02_distance_metrics

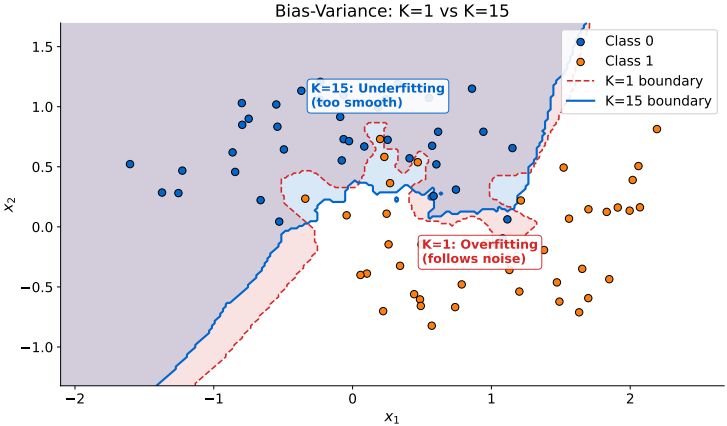
The choice of distance metric fundamentally shapes which neighbors are "nearest"

Why Feature Scaling Is Non-Negotiable



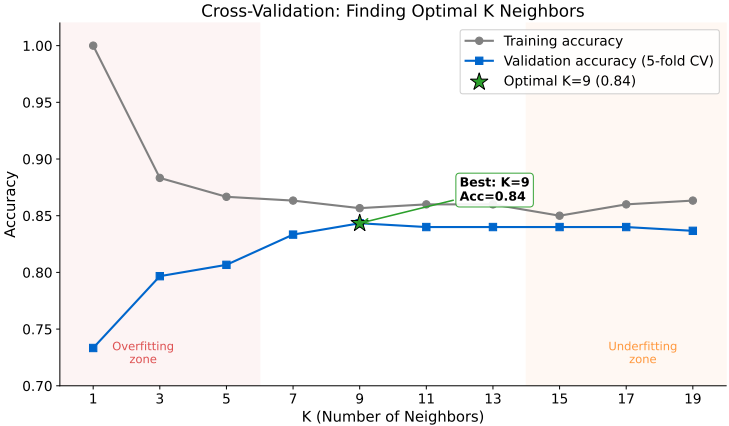
Unscaled features with larger ranges dominate distance calculations

The k Tradeoff: Bias vs Variance



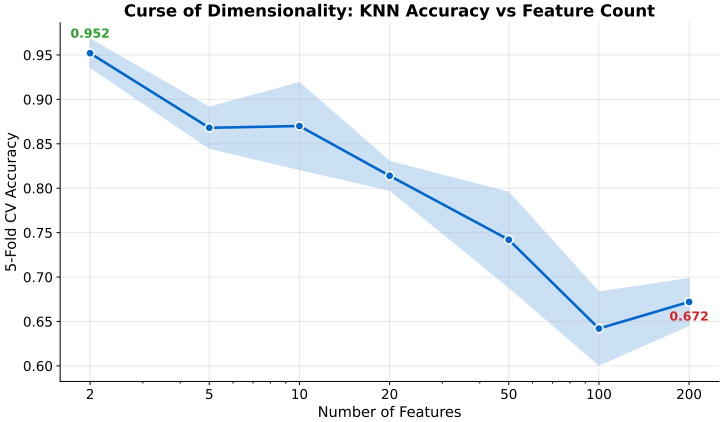
Small k = high variance (noisy), large k = high bias (over-smoothed)

Finding the Optimal k



Cross-validation reveals the sweet spot between underfitting and overfitting

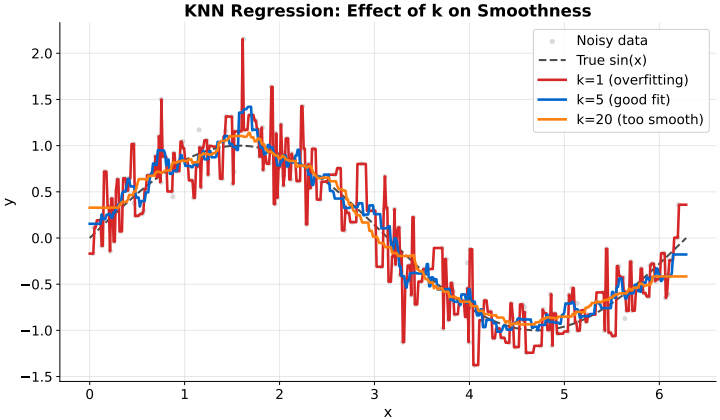
The Curse of Dimensionality



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L03_KNN_KMeans/top10_12_curse_of_dimensionality

As dimensions grow, all points become equidistant and KNN loses discriminative power

KNN for Regression: Averaging Neighbors



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L03_KNN_KMeans/top10_13_knn_regression

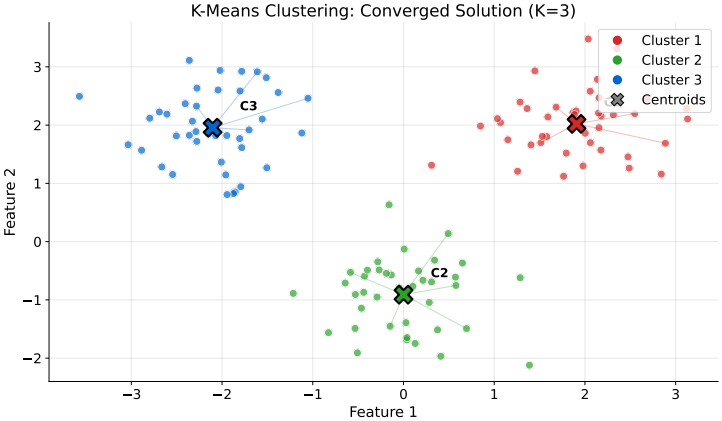
KNN regression averages neighbor values — k controls the smoothness of predictions

From Supervised to Unsupervised

- KNN needs labels — it **classifies** based on neighbor votes
- K-Means discovers structure — it **clusters** without labels
- Same distance intuition, fundamentally different tasks

Both algorithms rely on distance, but KNN is supervised while K-Means is unsupervised

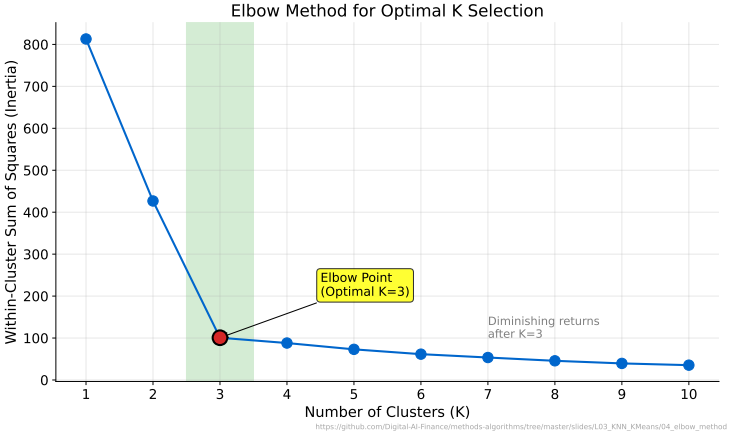
K-Means: The Iterative Algorithm



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L03_KNN_KMeans/03_kmeans_iteration

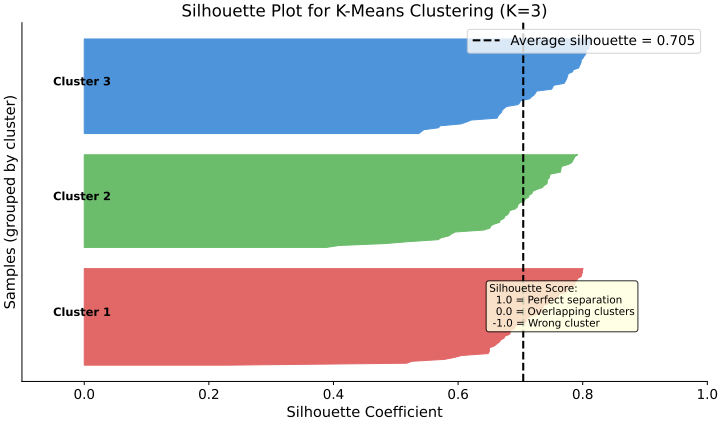
Each iteration alternates between assigning points and updating centroids

How Many Clusters? The Elbow Method



The "elbow" marks where adding clusters yields diminishing returns in inertia reduction

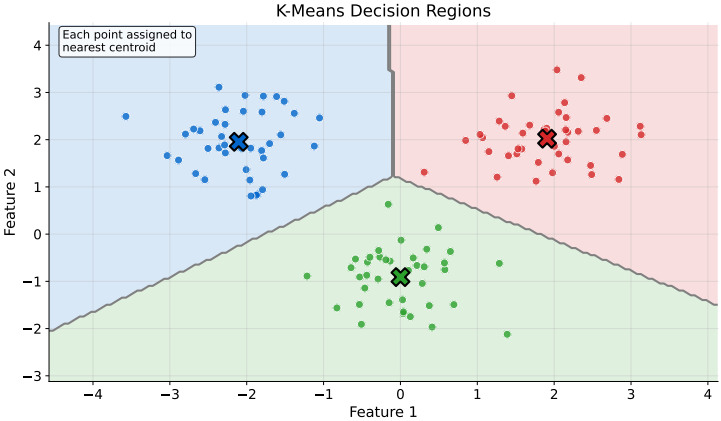
Silhouette Analysis: Cluster Quality



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L03_KNN_KMeans/05_silhouette

Silhouette scores measure how similar points are to their own cluster vs nearest neighbor cluster

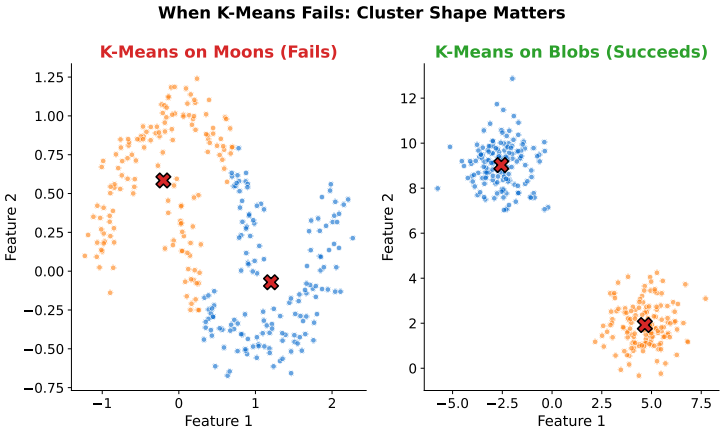
Voronoi Tessellation: K-Means Geometry



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L03_KNN_KMeans/06_voronoi

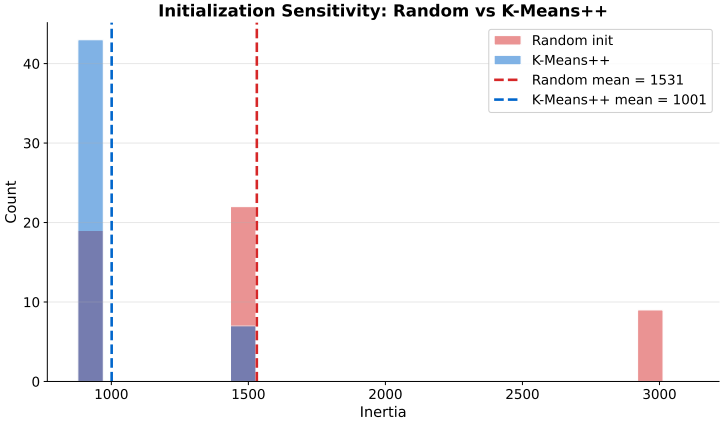
K-Means partitions space into Voronoi cells — each point belongs to its nearest centroid

When K-Means Fails: Cluster Shape Matters



K-Means assumes spherical clusters — non-convex shapes require density-based methods

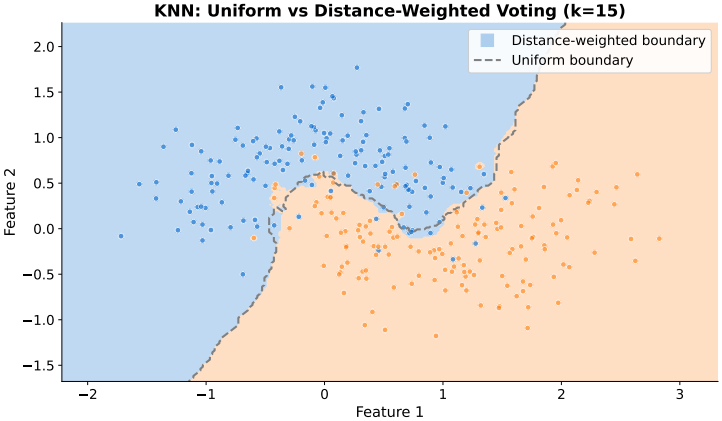
Initialization Matters: Random vs K-Means++



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L03_KNN_KMeans/top10_14_kmeans_init_sensitivity

K-Means++ provides smarter initialization, reducing sensitivity to starting positions

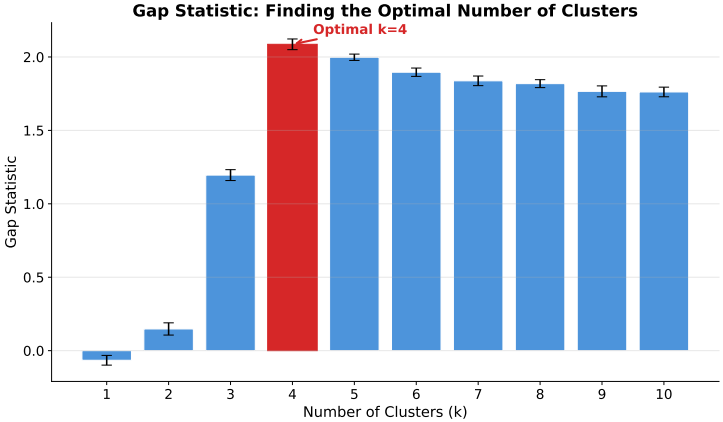
Distance-Weighted KNN: Closer Neighbors Matter More



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L03_KNN_KMeans/top10_11_knn_weighted_vs_uniform

Weighting by inverse distance reduces the influence of distant, possibly misleading neighbors

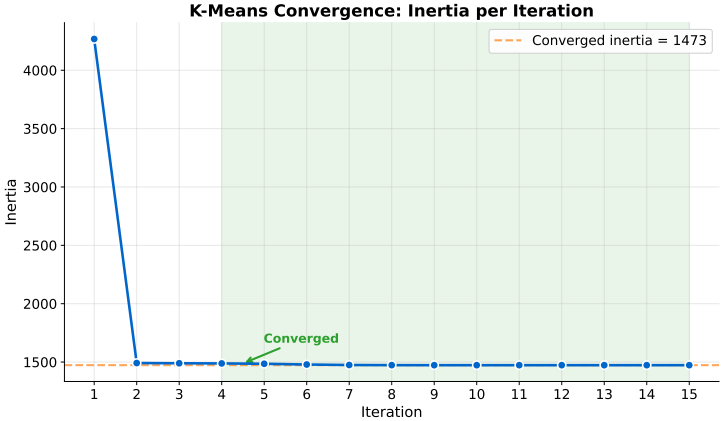
Gap Statistic: A Rigorous k Selection



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L03_KNN_KMeans/top10_17_gap_statistic

The gap statistic compares clustering quality against a null reference distribution

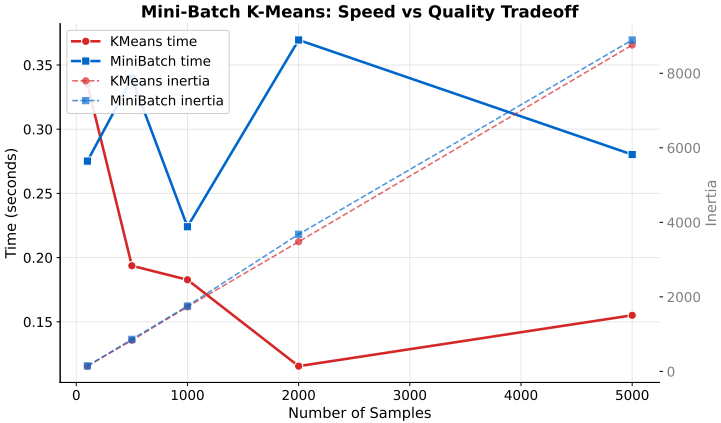
K-Means Convergence: Inertia per Iteration



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L03_KNN_KMeans/top10_19_kmeans_inertia_convergence

K-Means typically converges within 10–15 iterations for moderate-sized datasets

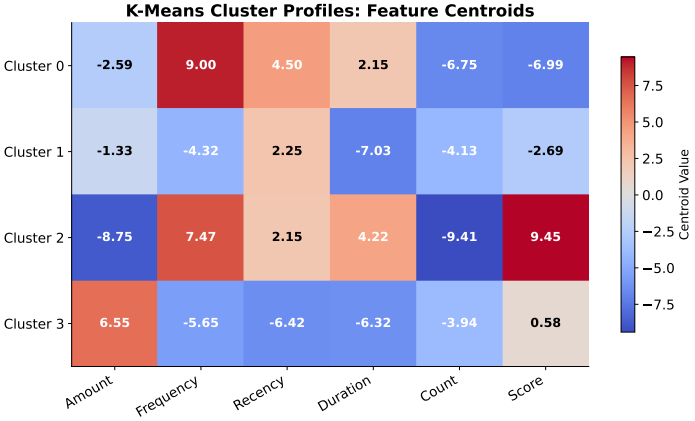
Mini-Batch K-Means: Speed vs Quality



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L03_KNN_KMeans/top10_15_minibatch_kmeans

Mini-batch updates trade marginal quality loss for dramatic speedup on large datasets

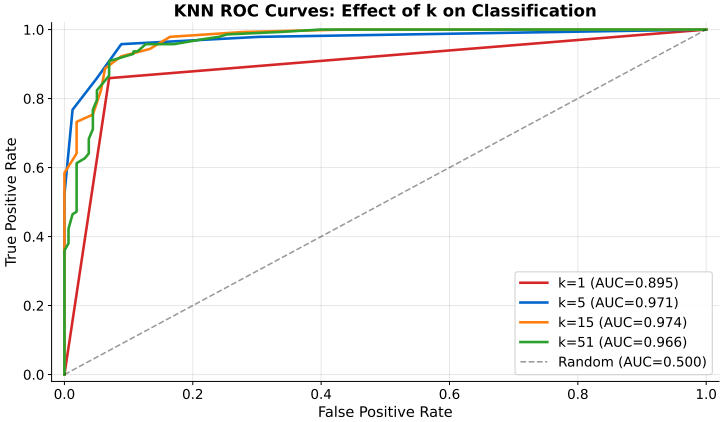
Cluster Profiles: What Each Segment Looks Like



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L03_KNN_KMeans/top10_16_cluster_centers_heatmap

Centroid feature values reveal the defining characteristics of each customer segment

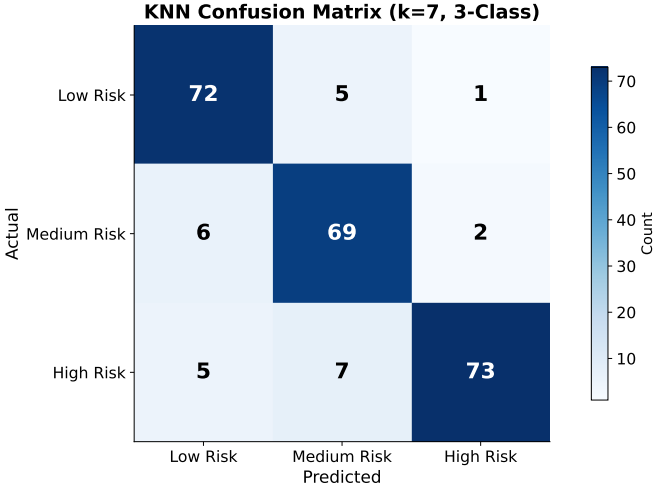
KNN ROC Curves: Effect of k on Classification



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L03_KNN_KMeans/top10_18_knn_roc_curves

Moderate k values typically achieve the best AUC by balancing precision and recall

KNN Confusion Matrix: Where Errors Happen



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L03_KNN_KMeans/top10_20_knn_confusion_matrix

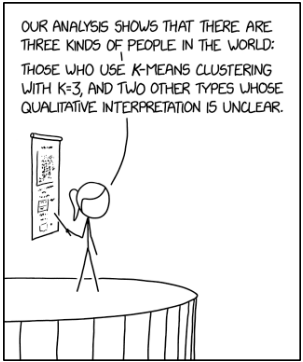
Confusion matrices reveal systematic misclassification patterns between classes

20 Charts You Should Know

1. Decision Boundary
2. Distance Metrics
3. Feature Scaling
4. Bias–Variance
5. CV Accuracy
6. Curse of Dimensionality
7. KNN Regression
8. Weighted vs Uniform
9. KNN ROC
10. KNN Confusion Matrix
11. K-Means Iterations
12. Elbow Method
13. Silhouette
14. Voronoi
15. Failure Cases
16. Init Sensitivity
17. Gap Statistic
18. Inertia Convergence
19. Mini-Batch
20. Cluster Profiles

Master these 20 visualizations for KNN and K-Means mastery

And Remember. . .



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